Course Orientation and the Probabilistic View of Advanced Machine Learning · Quiz explanations

Questions and compact answers reproduced from the instructor-provided Class 01 Notion archive. Full bilingual speaker scripts remain private.

1. What is the difference between $p(y \mid x)$ and $p(x \mid y)$?

They condition in opposite directions and are related by Bayes' rule, not by symmetry.

2. Why does independence imply zero covariance, while zero covariance does not generally imply independence?

Covariance captures only linear dependence; nonlinear dependence may remain.

3. Why is minimizing cross-entropy often equivalent to maximum likelihood estimation?

Cross-entropy is the NLL for Bernoulli or categorical observation models.

4. How does a prior become a regularization term in MAP estimation?

The negative log-prior is added to the NLL.

5. What is the difference between aleatoric and epistemic uncertainty?

Aleatoric uncertainty is intrinsic data variability; epistemic uncertainty reflects limited knowledge.

6. Why can a large learning rate cause divergence even for a convex objective?

The update can overshoot along directions of high curvature and oscillate or diverge.

7. What is lost when $p(\theta \mid \mathcal{D})$ is replaced by a point estimate $\hat{\theta}$?

Parameter uncertainty and model averaging are discarded, often producing overconfident predictions.